Affine-DDPNM: Methodology

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We extend the domain-decomposition pore-network method (DD-PNM) of [1] by enriching the interface traction space from a single constant-normal mode to a nine-mode affine representation. The method, termed **Affine-DDPNM**, retains the offline–online workflow and non-overlapping subdomain partition of the original DD-PNM. The only change is the interface Ansatz: on each internal face, the traction is now allowed to vary as an affine function of the in-plane coordinates, in both the normal and the two tangential directions.

1 Model problem

We consider steady, incompressible Stokes flow in a three-dimensional pore domain $\Omega \subset \mathbb{R}^3$. Let Γ_{wall} be the solid–fluid interface. A pressure drop $p_{\text{in}} - p_{\text{out}} > 0$ is imposed between the inlet and outlet boundaries, while all remaining external boundaries are treated as no-slip walls. Collecting all homogeneous Dirichlet segments in Γ_{dir}, the governing equations are

$$
\begin{aligned}
-\nu\Delta \boldsymbol{u} + \nabla p &= \mathbf{0}, && \text{in } \Omega, && (1)\\
\nabla \cdot \boldsymbol{u} &= 0, && \text{in } \Omega, && (2)\\
\boldsymbol{u} &= \mathbf{0}, && \text{on } \Gamma_{\text{dir}}, && (3)\\
\boldsymbol{n}\cdot\boldsymbol{\sigma}(\boldsymbol{u},p)\boldsymbol{n} &= -p_b, \quad (I - \boldsymbol{n}\otimes\boldsymbol{n})\,\boldsymbol{\sigma}(\boldsymbol{u},p)\boldsymbol{n} = \mathbf{0}, && \text{on } \Gamma_{\text{in}} \cup \Gamma_{\text{out}}, && (4)
\end{aligned}
$$

where $\boldsymbol{u}$ is velocity, p is pressure, ν is the kinematic viscosity, $\boldsymbol{n}$ is the outward unit normal, and $p_b = p_{\text{in}}$ on Γ_{in}, $p_b = p_{\text{out}}$ on Γ_{out}. The *pseudostress* tensor is

$$\boldsymbol{\sigma}(\boldsymbol{u},p) := -p\mathbf{I} + \nu\nabla\boldsymbol{u},$$

which for incompressible flow differs from the standard Cauchy stress $-p\mathbf{I} + 2\mu\boldsymbol{\varepsilon}(\boldsymbol{u})$ in the boundary traction, but the two coincide in the interior momentum equation. The zero tangential traction condition $(I - \boldsymbol{n}\otimes\boldsymbol{n})\,\boldsymbol{\sigma}(\boldsymbol{u},p)\boldsymbol{n} = \mathbf{0}$ is equivalent to $\boldsymbol{t}_\nu \cdot \boldsymbol{\sigma}(\boldsymbol{u},p)\boldsymbol{n} = 0$ for two orthonormal tangents $\boldsymbol{t}_1, \boldsymbol{t}_2$ spanning the boundary plane.

2 Domain partition

The pore domain Ω is partitioned into N non-overlapping Lipschitz subdomains $\{\Omega_i\}_{i=1}^N$, each corresponding to one pore body. For any pair (i,j) of adjacent pores, the shared interface is

$$\gamma_{ij} = \partial\Omega_i \cap \partial\Omega_j,$$

and we index the m internal interfaces as $\{\gamma_k\}_{k=1}^m$. Interfaces are assumed **planar** throughout the theoretical development; the numerical watershed experiments relax this and are discussed separately in Remark 2.

For each subdomain Ω_i we list its m_i ports as $\{\gamma_{i,1}, \ldots, \gamma_{i,m_i}\}$, partitioned into an unknown-traction set $\mathcal{U}_i$ (internal interfaces) and a known-traction set $\mathcal{K}_i$ (inlet/outlet boundaries). The local-to-global index maps and the Boolean restriction matrices R_i^u, R_i^k follow the convention of [1, Section 2.4].

The Voronoi partition places each interface as the plane perpendicular to and bisecting the line connecting the two neighbouring sphere centres; for unequal radii this plane differs from the exact clearance saddle surface. The geometric error introduced by this approximation is quantified in the numerical companion.

3 Reference finite-element discretization

Within each subdomain we employ the inf–sup stable Taylor–Hood pair $\mathbf{V}_h = [P_2]^3 \cap \mathbf{V}$, $Q_h = P_1 \cap Q$ on a body-fitted tetrahedral mesh, where

$$\mathbf{V} = \{\boldsymbol{v} \in [H^1(\Omega)]^3 : \boldsymbol{v} = \mathbf{0} \text{ on } \Gamma_{\text{dir}}\}, \qquad Q = L^2(\Omega).$$

With bases $\{\boldsymbol{\phi}_j\}_{j=1}^{n_u}$ and $\{\psi_l\}_{l=1}^{n_p}$, the discrete system on Ω_i under Neumann boundary data reads

$$\begin{bmatrix} A_i & B_i^\top \\ B_i & 0 \end{bmatrix} \begin{bmatrix} \mathbf{U}_i \\ \mathbf{P}_i \end{bmatrix} = \begin{bmatrix} \sum_{r=1}^{m_i} \boldsymbol{f}_i^{(r)} \\ \mathbf{0} \end{bmatrix}, \tag{5}$$

where A_i is the velocity-stiffness matrix (SPD), B_i is the divergence matrix, and $\boldsymbol{f}_i^{(r)}$ is the Neumann load vector assembled from the traction prescribed on port r. No pressure stabilisation is used ($\varepsilon = 0$).

4 Interface traction Ansatz

4.1 Classic DD-PNM (one mode per interface)

In the classic DD-PNM [1], the traction on each internal interface γ is uniform and purely normal:

$$-\boldsymbol{\sigma}(\boldsymbol{u}, p)\boldsymbol{n}_\gamma = p_\gamma \, \boldsymbol{n}_\gamma, \tag{6}$$

where $p_\gamma \in \mathbb{R}$ is a single scalar per interface. Equivalently, $\mathcal{T}_\gamma^{\mathrm{P0}} = \operatorname{span}\{\boldsymbol{n}_\gamma\}$.

4.2 Affine-DDPNM (nine modes per interface)

For each **planar** interface γ, we introduce a local orthonormal frame $\{\boldsymbol{n}_\gamma, \boldsymbol{t}_{\gamma,1}, \boldsymbol{t}_{\gamma,2}\}$ with origin at the interface centroid $\boldsymbol{c}_\gamma$. The dimensionless in-plane coordinates are

$$s(\boldsymbol{x}) = \frac{(\boldsymbol{x} - \boldsymbol{c}_\gamma) \cdot \boldsymbol{t}_{\gamma,1}}{a_\gamma}, \qquad t(\boldsymbol{x}) = \frac{(\boldsymbol{x} - \boldsymbol{c}_\gamma) \cdot \boldsymbol{t}_{\gamma,2}}{b_\gamma}, \tag{7}$$

where a_γ, b_γ are the half-spans of γ along the two tangent directions, ensuring $s, t = O(1)$ under shape-regularity.

For a pore Ω_i adjacent to γ, define the *signed* direction vectors

$$\boldsymbol{d}_{i,\gamma,\boldsymbol{n}} = \boldsymbol{n}_{i,\gamma}, \qquad \boldsymbol{d}_{i,\gamma,\boldsymbol{t}_\nu} = \sigma_{i,\gamma} \, \boldsymbol{t}_{\gamma,\nu} \;\; (\nu = 1, 2), \tag{8}$$

where $\sigma_{i,\gamma} = +1$ if Ω_i is the first member of the interface pair and -1 otherwise. These satisfy $\boldsymbol{d}_{j,\gamma,\beta} = -\boldsymbol{d}_{i,\gamma,\beta}$, guaranteeing that continuity conditions written below are sign-consistent.

The applied traction on γ (as seen from Ω_i) is expanded as

$$-\boldsymbol{\sigma}(\boldsymbol{u}, p) \, \boldsymbol{n}_{i,\gamma} = \sum_{(\alpha,\beta) \in \mathcal{M}} \lambda_{\gamma,\alpha,\beta} \, \varphi_\alpha(s,t) \, \boldsymbol{d}_{i,\gamma,\beta}, \tag{9}$$

where the **mode index set** is $\mathcal{M} = \{0, s, t\} \times \{\boldsymbol{n}, \boldsymbol{t}_1, \boldsymbol{t}_2\}$ with scalar shape functions $\varphi_0(s,t) = 1$, $\varphi_s(s,t) = s$, $\varphi_t(s,t) = t$. The minus sign on the left follows the classic convention $-\sigma n = p\, n$ and ensures that a positive $\lambda_{0,\boldsymbol{n}}$ corresponds to a pressure acting into the pore.

Each interface carries **nine** scalar coefficients $\boldsymbol{\lambda}_\gamma = (\lambda_{\gamma,0,\boldsymbol{n}},\ \lambda_{\gamma,0,\boldsymbol{t}_1},\ \lambda_{\gamma,0,\boldsymbol{t}_2},\ \lambda_{\gamma,s,\boldsymbol{n}},\ \ldots,\ \lambda_{\gamma,t,\boldsymbol{t}_2}) \in \mathbb{R}^9$ as the unknown degrees of freedom. The affine traction space is

$$\mathcal{T}_\gamma^{\text{Affine}} = \operatorname{span}\{\, \varphi_\alpha(s,t)\, \boldsymbol{d}_{i,\gamma,\beta} \mid (\alpha,\beta) \in \mathcal{M}\,\}.$$

Remark 1 (Behaviour of the nine modes)**.** *The constant normal mode $(\alpha,\beta) = (0,\boldsymbol{n})$ corresponds to a uniform normal pressure on the interface, matching the classic single-mode Ansatz. The linear normal modes $(s,\boldsymbol{n})$ and $(t,\boldsymbol{n})$ capture first-order spatial variation of the normal traction. The six tangential modes $(\alpha,\boldsymbol{t}_\nu)$ allow non-zero shear tractions at the interface, representing cross-flow and viscous momentum exchange that the purely normal classic Ansatz cannot express.*

Remark 2 (Planar interface requirement)**.** *The theory assumes planar interfaces with a fixed $\boldsymbol{n}_\gamma$. Watershed partitions produce bent, faceted interfaces where the mesh-level facet normal $\boldsymbol{n}(x)$ varies. In the code, watershed interfaces use the per-facet $\boldsymbol{n}(x)$ for the normal direction but a fixed averaged frame for the tangents; this deviates from the nine-mode space defined here and is one factor behind the degraded Affine accuracy on watershed geometries. All theoretical statements below apply to planar interfaces.*

5 Local compliance map

5.1 Primitive response loads

For each subdomain Ω_i and each mode $(\alpha,\beta) \in \mathcal{M}$ on an internal interface $\gamma_{i,r}$, the Neumann load vector is

$$\boldsymbol{f}_{i,r}^{(\alpha,\beta)} \in \mathbb{R}^{n_{i,u}}, \qquad \big[\boldsymbol{f}_{i,r}^{(\alpha,\beta)}\big]_j = \int_{\gamma_{i,r}} \varphi_\alpha(s,t)\, \boldsymbol{d}_{i,\gamma,\beta} \cdot \boldsymbol{\phi}_j^{(i)}\, \mathrm{d}S. \tag{10}$$

With the sign convention (9) $-\sigma n = \sum \lambda\varphi\boldsymbol{d}$, a positive coefficient λ produces a traction acting *into* the pore; the load vector thus carries no leading minus sign—it directly provides the right-hand side of (5) for the applied boundary force. For boundary ports only the uniform-normal mode $(\alpha,\beta) = (0,\boldsymbol{n})$ is used, with a *prescribed* coefficient equal to the inlet/outlet pressure.

Solving the local Stokes system for each unit load yields the velocity response $\mathbf{U}_{i,r}^{(\alpha,\beta)}$ and pressure response $\mathbf{X}_{i,r}^{(\alpha,\beta)}$:

$$\begin{bmatrix} A_i & B_i^\top \\ B_i & 0 \end{bmatrix} \begin{bmatrix} \mathbf{U}_{i,r}^{(\alpha,\beta)} \\ \mathbf{X}_{i,r}^{(\alpha,\beta)} \end{bmatrix} = \begin{bmatrix} \boldsymbol{f}_{i,r}^{(\alpha,\beta)} \\ \mathbf{0} \end{bmatrix}. \tag{11}$$

5.2 Compliance (Neumann-to-Dirichlet) map

Let $n_i = 9m_i^u + m_i^k$ be the total number of local traction coefficients (nine per internal interface, one per boundary port; the m_i^k boundary entries are *prescribed*, not unknown). Collect them in $\boldsymbol{\lambda}_i \in \mathbb{R}^{n_i}$.

By linearity of (5), the discrete velocity is

$$\mathbf{U}_i = \sum_{\gamma \in \mathcal{U}_i} \sum_{\alpha,\beta} \lambda_{i,\gamma,\alpha,\beta}\, \mathbf{U}_{i,\gamma}^{(\alpha,\beta)} \;+\; \sum_{\gamma \in \mathcal{K}_i} \lambda_{i,\gamma}^k\, \mathbf{U}_{i,\gamma}^{(0,\boldsymbol{n})}. \tag{12}$$

For each mode (α,β) on interface $\gamma_{i,r}$, the *weighted velocity moment* is

$$M_{i,r}^{(\alpha,\beta)} = \int_{\gamma_{i,r}} \varphi_\alpha(s,t)\, \boldsymbol{u}_{i,h} \cdot \boldsymbol{d}_{i,\gamma,\beta}\, \mathrm{d}S. \tag{13}$$

Only $M_{i,r}^{(0,\boldsymbol{n})} = \int_\gamma \boldsymbol{u} \cdot \boldsymbol{n} \, \mathrm{d}S$ is the volumetric flow rate. The remaining eight quantities are weighted velocity moments: $M^{(s,\boldsymbol{n})}, M^{(t,\boldsymbol{n})}$ are linear normal flux moments, and $M^{(\alpha,\boldsymbol{t}_1)}, M^{(\alpha,\boldsymbol{t}_2)}$ are tangential velocity moments. Collectively we call them the **nine weighted velocity moments**.

The local compliance (Neumann-to-Dirichlet) map $\mathbf{G}_i \in \mathbb{R}^{n_i \times n_i}$ has entries

$$\mathbf{G}_i\big[(\gamma,\alpha,\beta),\,(\gamma',\alpha',\beta')\big] = \big[\boldsymbol{f}_{i,\gamma}^{(\alpha,\beta)}\big]^\top \mathbf{U}_{i,\gamma'}^{(\alpha',\beta')}. \tag{14}$$

For boundary ports, only the $(\alpha,\beta) = (0,\boldsymbol{n})$ entry is stored. The map is symmetric positive semi-definite; the energy identity and kernel characterisation are given in the companion mathematical properties document.

$\mathbf{G}_i$ is block-partitioned into $\mathbf{G}_i^{uu}$ (unknown–unknown) and $\mathbf{G}_i^{uk}$ (unknown–known) as in the classic case.

6 Global Schur complement assembly

Let $\boldsymbol{\lambda} \in \mathbb{R}^{9m}$ collect all unknown interface coefficients and $\boldsymbol{\lambda}^k \in \mathbb{R}^{m^k}$ the prescribed boundary tractions. Enforcing continuity of **all nine weighted velocity moments** on every internal interface yields the global Schur system

$$\mathbf{S}\,\boldsymbol{\lambda} = \boldsymbol{F}, \qquad \mathbf{S} = \sum_{i=1}^{N} (R_i^u)^\top \mathbf{G}_i^{uu} R_i^u, \qquad \boldsymbol{F} = -\sum_{i=1}^{N} (R_i^u)^\top \mathbf{G}_i^{uk} R_i^k \boldsymbol{\lambda}^k, \tag{15}$$

where $\mathbf{S} \in \mathbb{R}^{9m \times 9m}$. The continuity condition $M_{i,r}^{(\alpha,\beta)} + M_{j,r}^{(\alpha,\beta)} = 0$ follows from $\boldsymbol{d}_{j,\gamma,\beta} = -\boldsymbol{d}_{i,\gamma,\beta}$ and is imposed for all nine modes, not only the constant normal component.

Remark 3 (SPD of the global system)**.** $\mathbf{S}$ *is symmetric by construction. Positive definiteness requires:*

1. *Every connected component of the pore adjacency graph contains at least one pore with a prescribed-traction (inlet/outlet) port (the* anchoring condition*);*
2. *Each pore has a positive-measure no-slip wall segment, fixing rigid-body modes;*
3. *The nine primitive loads per interface are linearly independent on each face (verified numerically for all tested geometries);*
4. *Interface coordinate frames and side signs are consistent across adjacent pores.*

Under these conditions, the Schur system is SPD and uniquely solvable. In practice, watershed partitions may produce floating basins *(pores with no solid-wall facet); these are merged into their neighbours before the Schur assembly to enforce the anchoring condition.*

7 Solution algorithm

The Affine-DDPNM workflow is summarised in Algorithm 7. The differences from the classic algorithm are: (i) nine primitive loads per internal interface instead of one, and (ii) the 9× increase in the size of the local compliance blocks and the global Schur system. The offline–online decomposition and the embarrassingly parallel structure are preserved unchanged.

Algorithm 7: Affine-DDPNM solver (assemble – couple – recover)

Input: subdomain meshes, FE pairs $(V_{i,h}, Q_{i,h})$, prescribed tractions $\boldsymbol{\lambda}^k$.

Output: unknown interface coefficients $\boldsymbol{\lambda}$, weighted velocity moments, and (optionally) local fields.

1. **Offline library** — For each pore body $i = 1, \dots, N$ (in parallel):
   (a) Assemble local operators A_i, B_i.
   (b) For each internal interface $\gamma \in \mathcal{U}_i$ and each mode $(\alpha, \beta) \in \mathcal{M}$: assemble unit load $\boldsymbol{f}_{i,\gamma}^{(\alpha,\beta)}$ via (10), solve (11) for $(\mathbf{U}_{i,\gamma}^{(\alpha,\beta)}, \mathbf{X}_{i,\gamma}^{(\alpha,\beta)})$.
   (c) For each boundary port $\gamma \in \mathcal{K}_i$: assemble and solve the single uniform-normal load $(\alpha, \beta) = (0, \boldsymbol{n})$.
   (d) Form local compliance block $\mathbf{G}_i$ via (14).
   (e) Accumulate $S \mathrel{+}= (R_i^u)^\top \mathbf{G}_i^{uu} R_i^u$, $\boldsymbol{F} \mathrel{-}= (R_i^u)^\top \mathbf{G}_i^{uk} R_i^k \boldsymbol{\lambda}^k$.
2. **Global solve** — Solve $\mathbf{S}\,\boldsymbol{\lambda} = \boldsymbol{F}$ by sparse direct solver.
3. **Reconstruction** (optional, in parallel):

$$\mathbf{U}_i = \sum_{\gamma \in \mathcal{U}_i} \sum_{\alpha,\beta} \lambda_{i,\gamma,\alpha,\beta}\, \mathbf{U}_{i,\gamma}^{(\alpha,\beta)} + \sum_{\gamma \in \mathcal{K}_i} \lambda_{i,\gamma}^k\, \mathbf{U}_{i,\gamma}^{(0,\boldsymbol{n})},$$

$$\mathbf{P}_i = \sum_{\gamma \in \mathcal{U}_i} \sum_{\alpha,\beta} \lambda_{i,\gamma,\alpha,\beta}\, \mathbf{X}_{i,\gamma}^{(\alpha,\beta)} + \sum_{\gamma \in \mathcal{K}_i} \lambda_{i,\gamma}^k\, \mathbf{X}_{i,\gamma}^{(0,\boldsymbol{n})}.$$

8 Relation to the classic DD-PNM

Let $\mathcal{P}_{\mathrm{P0}} : \mathbb{R}^m \to \mathbb{R}^{9m}$ be the embedding that maps the m classic constant-normal coefficients to the $9m$ affine space by placing them in the $(0, \boldsymbol{n})$ slots and setting all other components to zero. The classic traction space $\mathrm{Range}(\mathcal{P}_{\mathrm{P0}})$ is a subspace of the affine traction space.

The classic DD-PNM is recovered by **restricting both the trial and test spaces** to the constant-normal subspace:

$$\mathbf{S}_{\mathrm{P0}} = \mathcal{P}_{\mathrm{P0}}^\top \mathbf{S}\, \mathcal{P}_{\mathrm{P0}}, \qquad \boldsymbol{F}_{\mathrm{P0}} = \mathcal{P}_{\mathrm{P0}}^\top \boldsymbol{F}, \tag{16}$$

yielding the classic $m \times m$ Schur system. In general, the classic solution $\boldsymbol{p} = \mathbf{S}_{\mathrm{P0}}^{-1} \boldsymbol{F}_{\mathrm{P0}}$ does *not* coincide with the $(0, \boldsymbol{n})$-components of the full $9m$-system solution $\boldsymbol{\lambda} = \mathbf{S}^{-1}\boldsymbol{F}$, because the eight additional test-space constraints alter the Galerkin projection.

The enrichment guarantee is that the affine space contains the classic space as a subspace of the traction representation, and the Galerkin optimality (proved in the companion mathematical properties document) ensures that the affine error in the $\widehat{\mathbf{S}}$-energy norm is never larger than the classic error. The 73–90% L^2 reductions observed numerically are *consistent* with this energy-norm guarantee.

9 Computational cost

For a pore with m_i ports, the local compliance matrix $\mathbf{G}_i$ has size $n_i \times n_i$ where $n_i \approx 9m_i$ (versus m_i in the classic method). Building $\mathbf{G}_i$ requires n_i independent local Stokes solves; this offline work is embarrassingly parallel. The global Schur system $\mathbf{S}$ has size $9m \times 9m$ and is solved by a sparse direct solver.

Offline cost grows approximately $9\times$ in the number of local solves. Online cost is dominated by the sparse LU factorisation of $\mathbf{S}$, which has at most $9 \times (\text{max coordination}) \times 9$ non-zero blocks per row. For multi-boundary-condition studies the offline library is built once and reused; the on-line solve then achieves the speedups reported in the numerical companion (100–$10^4\times$ over monolithic FEM for Classic, 1.7–840$\times$ for Affine).

References

[1] S. Sun, Z. Wang, L. Zhang, J. Zhao. A domain decomposition approach to pore-network modeling of porous media flow. *arXiv:2510.13429v2*, 2026.